Fundamentals of Physics

Tashi Wangchuk

Druk Gyalpo's Institute, Pangbisa, Paro 12001, Bhutan

tashiwangchuk@academy.bt

April 2026

Acknowledgements

These lecture notes are anything but original. My main contribution has been to gather, adapt, and weave together the clearest discussions and explanations I could find in the vast literature on the subject. The lectures lean heavily on the sources listed in the syllabus, and much of the brainstorming was done in conversation with Claude AI.

I thank my students, whose evident grasp of the material has shown me where to simplify and where to linger in detail. I am supported by the Druk Gyalpo's Institute.

Table of Contents

CHAPTER 1

Vectors

1.1 Introduction to Scalars and Vectors

Imagine you walk from corner A to corner C (a distance of 4 m), then turn and walk from C to B (a distance of 3 m).

- The total *distance* you walked is $4 + 3 = 7\,\mathrm{m}$. Distance only cares about how much, not which way.
- Your *displacement* — the straight-line shortcut from start to finish — is only 5 m, pointing from A to B. Displacement cares about how much *and* which way.

Scalars and Vectors

A **scalar** is a quantity that has magnitude only (no direction): distance, mass, time, temperature, speed, energy.

A **vector** is a quantity that has magnitude and direction: displacement, velocity, acceleration, force.

	Scalar	Vector
Has magnitude	yes	yes
Has direction	no	yes
Examples	distance, speed, mass, time	displacement, velocity, force

Notation

- Vectors are written with a small arrow above the symbol, e.g. a force $\vec{F}$ or a velocity $\vec{v}$.
- The magnitude (size) of a vector $\vec{A}$ is written $|\vec{A}|$, or simply A.
- In diagrams a vector is drawn as an arrow: the length of the arrow is proportional to the magnitude, and the arrowhead shows the direction.

Vectors have magnitude and direction, but not location

A displacement of 1 m due east from Paro is represented by the *same* vector as a displacement of 1 m due east from Thimphu. Sliding a vector to a new position (without rotating or resizing it) does not change the vector at all.

The negative of a vector

$-\vec{A}$ is a vector with the same magnitude as $\vec{A}$ but pointing in the opposite direction.

1.2 Vector Operations

Addition of vectors (head-to-tail / triangle rule)

To add two vectors, place the tail of $\vec{B}$ at the head of $\vec{A}$. The sum $\vec{A}+\vec{B}$ is the vector drawn from the tail of $\vec{A}$ to the head of $\vec{B}$. This sum is called the **resultant vector**.

Addition is **commutative**: $\vec{A}+\vec{B}=\vec{B}+\vec{A}$.

Addition is **associative**: $(\vec{A}+\vec{B})+\vec{C}=\vec{A}+(\vec{B}+\vec{C})$.

Subtraction means adding the opposite: $\vec{A}-\vec{B}=\vec{A}+(-\vec{B})$.

Parallelogram law of vector addition

Draw $\vec{a}$ and $\vec{b}$ with both tails together. Treat them as two adjacent sides and complete the parallelogram. The diagonal through the common tails is the sum $\vec{a}+\vec{b}=\vec{R}$.

Deriving the magnitude and direction of the resultant. Let $a=|\vec{a}|$, $b=|\vec{b}|$, and let θ be the angle between them. By the law of cosines applied to the triangle formed:

$$R^2 = a^2 + 2ab\cos\theta + b^2,$$

so

Magnitude and direction of the resultant

$$R = |\vec{a}+\vec{b}| = \sqrt{a^2 + 2ab\cos\theta + b^2}$$

$$\alpha = \tan^{-1}\left(\frac{b\sin\theta}{a+b\cos\theta}\right) \qquad \text{(angle that } \vec{R} \text{ makes with } \vec{a}\text{)}$$

Worked Example 1.1: Resultant of two forces

Two forces of $a = 3\,\mathrm{N}$ and $b = 4\,\mathrm{N}$ act at a point with an angle of $\theta = 60^\circ$ between them. Find the magnitude and direction of the resultant.

Solution.

$$R = \sqrt{3^2 + 2(3)(4)\cos 60^\circ + 4^2} = \sqrt{9+12+16} = \sqrt{37} \approx 6.1 \text{ N}.$$

$$\alpha = \tan^{-1}\left(\frac{4\sin 60^\circ}{3+4\cos 60^\circ}\right) = \tan^{-1}\left(\frac{3.46}{5}\right) \approx 34.7^\circ$$

measured from the $3\,\mathrm{N}$ force.

Try It Yourself 1.1: Two velocities

A boat heads north at $a = 6\,\text{m/s}$ while a current pushes it at $b = 8\,\text{m/s}$ at $\theta = 90°$ to the boat's heading. Show that the resultant speed is $10\,\text{m/s}$ and find its direction relative to north.

Multiplication by a scalar

Multiplying a vector by a positive scalar n multiplies its magnitude but leaves the direction unchanged. If n is negative, the direction is reversed. Scalar multiplication is distributive:

$$n(\vec{A} + \vec{B}) = n\vec{A} + n\vec{B}.$$

The dot (scalar) product

Dot product

$$\vec{A} \cdot \vec{B} \equiv AB\cos\theta,$$

where θ is the angle between the vectors when placed tail to tail. The result is a scalar — hence the alternative name *scalar product*.

Geometrically, $\vec{A} \cdot \vec{B}$ is the magnitude of $\vec{A}$ times the projection of $\vec{B}$ along $\vec{A}$ (which is $B\cos\theta$), or equivalently B times the projection of $\vec{A}$ along $\vec{B}$.

Properties.

- Commutative: $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$.
- Distributive: $\vec{A} \cdot (\vec{B} + \vec{C}) = \vec{A} \cdot \vec{B} + \vec{A} \cdot \vec{C}$.
- A vector dotted with itself gives its magnitude squared: $\vec{A} \cdot \vec{A} = A^2$.
- If $\vec{A} \perp \vec{B}$, then $\vec{A} \cdot \vec{B} = AB\cos 90° = 0$.

Worked Example 1.2: Work done by a force

A force $\vec{F}$ of magnitude $10\,\text{N}$ acts at $30°$ to a displacement $\vec{d}$ of magnitude $5\,\text{m}$. The work done is $W = \vec{F} \cdot \vec{d}$:

$$W = Fd\cos\theta = (10)(5)\cos 30° \approx 43.3 \text{ J}.$$

Worked Example 1.3: Class exercise — the Law of Cosines

Let $\vec{C} = \vec{A} - \vec{B}$. Calculate the dot product of $\vec{C}$ with itself.

Solution.

$$\begin{aligned}\vec{C} \cdot \vec{C} &= (\vec{A} - \vec{B}) \cdot (\vec{A} - \vec{B}) = \vec{A} \cdot \vec{A} - \vec{A} \cdot \vec{B} - \vec{B} \cdot \vec{A} + \vec{B} \cdot \vec{B} \\ &= A^2 - AB\cos\theta - BA\cos\theta + B^2.\end{aligned}$$

$$\boxed{C^2 = A^2 + B^2 - 2AB\cos\theta} \qquad \text{(the Law of Cosines).}$$

The cross (vector) product

Cross product

$$\vec{A} \times \vec{B} \equiv AB\sin\theta\,\hat{n},$$

where $\hat{n}$ is a unit vector pointing perpendicular to the plane containing $\vec{A}$ and $\vec{B}$. The result is itself a vector — hence the name *vector product*.

The magnitude is $|\vec{A} \times \vec{B}| = AB\sin\theta$.

Right-hand rule

Point the fingers of your right hand along the first vector, curl them through the smaller angle toward the second vector — your thumb then points along $\hat{n}$.

Properties.

- Distributive: $\vec{A} \times (\vec{B} + \vec{C}) = \vec{A} \times \vec{B} + \vec{A} \times \vec{C}$.
- **Not commutative** — order matters: $\vec{B} \times \vec{A} = -(\vec{A} \times \vec{B})$.
- Geometrically, $|\vec{A} \times \vec{B}|$ equals the area of the parallelogram with sides $\vec{A}$ and $\vec{B}$.
- Parallel vectors give zero: $\vec{A} \times \vec{A} = A^2 \sin 0^\circ\,\hat{n} = \vec{0}$.

Worked Example 1.4: Torque

A spanner is turned by a force $F = 20\,\mathrm{N}$ applied at the end of a handle of length $r = 0.30\,\mathrm{m}$, at 90° to the handle. The torque magnitude is

$$|\vec{\tau}| = |\vec{r} \times \vec{F}| = rF\sin 90^\circ = (0.30)(20)(1) = 6\ \mathrm{N\,m}.$$

1.3 Components of a Vector

It is often useful to describe a vector in terms of its components. If a vector $\vec{A}$ is split into two mutually perpendicular vectors, these are called the **rectangular components** of $\vec{A}$.

The vector $\vec{A}$ has magnitude A and makes angle θ with the positive x-axis.

Finding the components from magnitude and angle

$$A_x = A\cos\theta, \qquad A_y = A\sin\theta.$$

We can reverse the process to recover the magnitude and direction:

$$A_x^2 + A_y^2 = A^2(\cos^2\theta + \sin^2\theta) = A^2, \qquad \frac{A_y}{A_x} = \tan\theta.$$

Finding magnitude and angle from the components

$$A = \sqrt{A_x^2 + A_y^2}, \qquad \theta = \tan^{-1}\left(\frac{A_y}{A_x}\right).$$

Worked Example 1.5: Resolving a force

A force of $A = 100\,\mathrm{N}$ acts at $\theta = 30°$ above the horizontal. Its components are

$$A_x = 100\cos 30° \approx 86.6\ \mathrm{N}, \qquad A_y = 100\sin 30° = 50\ \mathrm{N}.$$

Worked Example 1.6: Adding vectors using components

Add $\vec{A}$ ($5\,\mathrm{m}$ at $37°$) and $\vec{B}$ ($8\,\mathrm{m}$ at $0°$, i.e. along x).

Solution. Resolve each, add like components, then recombine.

$$\begin{aligned} A_x &= 5\cos 37° \approx 4.0, & A_y &= 5\sin 37° \approx 3.0,\\ B_x &= 8, & B_y &= 0. \end{aligned}$$

$$R_x = 4.0 + 8 = 12.0, \qquad R_y = 3.0 + 0 = 3.0.$$

$$R = \sqrt{12.0^2 + 3.0^2} \approx 12.4\ \mathrm{m}, \qquad \theta = \tan^{-1}\left(\frac{3.0}{12.0}\right) \approx 14°.$$

Try It Yourself 1.2: Components of velocity

A projectile is launched at $20\,\mathrm{m/s}$, $60°$ above the horizontal. Find the horizontal and vertical components of the launch velocity.

(Answer: $v_x = 10\ m/s$, $v_y \approx 17.3\ m/s$.)

1.4 Vector Algebra in Components

In practice it is often easiest to set up Cartesian coordinates x, y, z and work with components.

Basis vectors and unit vectors

Let $\hat{x}, \hat{y}, \hat{z}$ (sometimes written $\hat{\imath}, \hat{\jmath}, \hat{k}$) be unit vectors pointing along the x-, y-, and z-axes. These are the **basis vectors**.

Unit vector

A unit vector has magnitude exactly 1; its only job is to indicate direction. Any vector $\vec{A}$ can be turned into a unit vector (written $\hat{A}$) by dividing by its own magnitude:

$$\hat{A} = \frac{\vec{A}}{|\vec{A}|}.$$

Component form of a vector

Any vector can be written in terms of the basis vectors:

$$\vec{A} = A_x\hat{x} + A_y\hat{y} + A_z\hat{z}.$$

The coefficients A_x, A_y, A_z are the *components* of $\vec{A}$ — geometrically, the projections of $\vec{A}$ onto the three axes. A component can be extracted by dotting with the matching basis vector:

$$\vec{A}\cdot\hat{x} = A_x, \qquad \vec{A}\cdot\hat{y} = A_y, \qquad \vec{A}\cdot\hat{z} = A_z.$$

The four operations in component form

Rule 1 — Addition: add like components

$$\vec{A} + \vec{B} = (A_x + B_x)\hat{x} + (A_y + B_y)\hat{y} + (A_z + B_z)\hat{z}$$

Rule 2 — Scalar multiplication: scale every component

$$a\vec{A} = (aA_x)\hat{x} + (aA_y)\hat{y} + (aA_z)\hat{z}$$

Rule 3 — Dot product: multiply like components and add

$$\vec{A}\cdot\vec{B} = A_xB_x + A_yB_y + A_zB_z$$

This works because the basis vectors are mutually perpendicular unit vectors:

$$\hat{x}\cdot\hat{x} = \hat{y}\cdot\hat{y} = \hat{z}\cdot\hat{z} = 1, \qquad \hat{x}\cdot\hat{y} = \hat{x}\cdot\hat{z} = \hat{y}\cdot\hat{z} = 0.$$

In particular $\vec{A}\cdot\vec{A} = A_x^2 + A_y^2 + A_z^2$, so

Magnitude (3-D Pythagoras)

$$|\vec{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

Bonus — Rule 4: the cross product as a determinant

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} = (A_y B_z - A_z B_y)\hat{x} - (A_x B_z - A_z B_x)\hat{y} + (A_x B_y - A_y B_x)\hat{z}$$

Worked Example 1.7: Putting the rules together

Let $\vec{A} = 2\hat{x} + 3\hat{y} + \hat{z}$ and $\vec{B} = \hat{x} - \hat{y} + 4\hat{z}$.

(a) *Sum.* $\vec{A} + \vec{B} = (2+1)\hat{x} + (3-1)\hat{y} + (1+4)\hat{z} = 3\hat{x} + 2\hat{y} + 5\hat{z}$.

(b) *Dot product.* $\vec{A} \cdot \vec{B} = (2)(1) + (3)(-1) + (1)(4) = 2 - 3 + 4 = 3$.

(c) *Magnitude of $\vec{A}$.* $|\vec{A}| = \sqrt{2^2 + 3^2 + 1^2} = \sqrt{14} \approx 3.74$.

(d) *Angle between them.* Using $\vec{A} \cdot \vec{B} = AB\cos\theta$ with $|\vec{B}| = \sqrt{1+1+16} = \sqrt{18}$:

$$\cos\theta = \frac{3}{\sqrt{14}\sqrt{18}} \approx 0.189 \implies \theta \approx 79^\circ.$$

1.5 Position, Displacement, and Separation Vectors

Position vector

A point in space is described by its Cartesian coordinates: (x, y) in 2-D or (x, y, z) in 3-D. The vector from the origin O to that point is the **position vector**:

$$\vec{r} \equiv x\hat{x} + y\hat{y} + z\hat{z}.$$

Its magnitude is the distance from the origin, $r = \sqrt{x^2 + y^2 + z^2}$, and the unit vector pointing radially outward is

$$\hat{r} = \frac{\vec{r}}{r} = \frac{x\hat{x} + y\hat{y} + z\hat{z}}{\sqrt{x^2 + y^2 + z^2}}.$$

Separation vector

Separation vector $\vec{\Delta}$

The separation vector is the displacement from a source point $\vec{r}\,'$ (where something is located, e.g. an electric charge) to a field point $\vec{r}$ (where you are observing):

$$\vec{\Delta} \equiv \vec{r} - \vec{r}\,'.$$

Its magnitude is $\Delta = |\vec{r} - \vec{r}\,'|$, and the unit vector pointing from the source to the field point is

$$\hat{\Delta} = \frac{\vec{\Delta}}{\Delta} = \frac{\vec{r} - \vec{r}\,'}{|\vec{r} - \vec{r}\,'|}.$$

In Cartesian coordinates,

$$\vec{\Delta} = (x - x')\hat{x} + (y - y')\hat{y} + (z - z')\hat{z},$$
$$\Delta = \sqrt{(x - x')^2 + (y - y')^2 + (z - z')^2},$$
$$\hat{\Delta} = \frac{(x - x')\hat{x} + (y - y')\hat{y} + (z - z')\hat{z}}{\sqrt{(x - x')^2 + (y - y')^2 + (z - z')^2}}.$$

Worked Example 1.8: A separation vector

A charge sits at the source point $\vec{r}\,' = (1, 0, 2)$ and we observe at the field point $\vec{r} = (4, 4, 2)$ (coordinates in metres). Then

$$\vec{\Delta} = (4 - 1)\hat{x} + (4 - 0)\hat{y} + (2 - 2)\hat{z} = 3\hat{x} + 4\hat{y},$$

$$\Delta = \sqrt{3^2 + 4^2 + 0^2} = 5 \text{ m}, \qquad \hat{\Delta} = \tfrac{3}{5}\hat{x} + \tfrac{4}{5}\hat{y}.$$

When are two vectors equal?

Two vectors are equal only if their corresponding components are equal — equivalently, if they have the same magnitude and the same direction. Sliding a vector to a new position (a parallel translation) does not change it.

Formula Summary: Vectors

Formula Summary: Vectors

Quantity	Formula
Resultant (parallelogram law)	$R = \sqrt{a^2 + 2ab\cos\theta + b^2}$
Direction of resultant	$\alpha = \tan^{-1}\left(\frac{b\sin\theta}{a+b\cos\theta}\right)$
Dot product	$\vec{A}\cdot\vec{B} = AB\cos\theta = A_xB_x + A_yB_y + A_zB_z$
Cross-product magnitude	$\lvert\vec{A}\times\vec{B}\rvert = AB\sin\theta$
Components (2-D)	$A_x = A\cos\theta, \quad A_y = A\sin\theta$
Magnitude (2-D)	$A = \sqrt{A_x^2 + A_y^2}$
Magnitude (3-D)	$A = \sqrt{A_x^2 + A_y^2 + A_z^2}$
Unit vector	$\hat{A} = \vec{A}/\lvert\vec{A}\rvert$
Position vector	$\vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$
Separation vector	$\vec{\Delta} = \vec{r} - \vec{r}\,'$

CHAPTER 2

Dimensional Analysis

2.1 Dimension

In physics, we measure many different things, but they are all built from a small set of seven fundamental base quantities (see Table 2.1).

Table 2.1: The seven fundamental base quantities, their dimensions, and sample units.

Base Quantity	Dimension	Sample Units
Mass	$[M]$	kg, g, lbs
Length	$[L]$	m, cm, feet, miles
Time	$[T]$	s, minutes, hours, years
Electric Current	$[I]$	A, mA
Thermodynamic Temperature	$[\Theta]$	K, °C, °F
Amount of Substance	$[N]$	mol
Luminous Intensity	$[J]$	cd

The **dimension** of any physical quantity represents the powers to which these base quantities must be raised to express that quantity. Example:

$$[\text{Force}] = [\text{mass}] \times \frac{[\text{velocity}]}{[\text{time}]} = [\text{mass}] \times \frac{[\text{length/time}]}{[\text{time}]} = MLT^{-2}.$$

Such an expression for a physical quantity in terms of the base quantities is called the **dimensional formula**. Thus, the dimensional formula of force is MLT^{-2}. In other words, we say the dimensions of force are 1 in mass, 1 in length, and -2 in time. The dimensions in all other base quantities are zero.

Note that in this type of calculation the magnitudes are not considered. It is equality of the *type* of quantity that enters. Thus, change in velocity, initial velocity, average velocity, and final velocity are all equivalent in this discussion — each one is LT^{-1}.

The physical quantity that is expressed in terms of the base quantities is enclosed in square brackets to remind us that the equation is among the dimensions and not among the magnitudes.

2.2 The Principle of Homogeneity

You cannot add apples to oranges, and in physics, you cannot add velocity to force.

The Principle of Homogeneity

The dimensions of all separated terms in a physical equation must be identical.

Let us check the kinematic equation $x = ut + \frac{1}{2}at^2$:

- $[x] = L$
- $[ut] = (LT^{-1})(T) = L$
- $[\frac{1}{2}at^2] = (LT^{-2})(T^2) = L$

Since every term evaluates to length $[L]$, the equation is dimensionally correct. Note: the constant $\frac{1}{2}$ has no dimension and is ignored in this check.

2.3 Uses of Dimensional Analysis

Checking the Correctness of an Equation

As shown above, if an equation is not dimensionally homogeneous, it is physically impossible and strictly wrong.

Limitation: A dimensionally correct equation might still be wrong if numerical constants are missing.

Conversion of Units

Dimensions help us find conversion factors between different unit systems (like SI and CGS). For example, to convert 1 Pascal (SI unit of pressure) to CGS pressure:

$$[\text{Pressure}] = \frac{[\text{Force}]}{[\text{Area}]} = \frac{MLT^{-2}}{L^2} = ML^{-1}T^{-2}.$$

$$1 \text{ Pascal} = (1 \text{ kg})(1 \text{ m})^{-1}(1 \text{ s})^{-2}, \qquad 1 \text{ CGS pressure} = (1 \text{ g})(1 \text{ cm})^{-1}(1 \text{ s})^{-2}.$$

$$\frac{1 \text{ Pascal}}{1 \text{ CGS pressure}} = \left(\frac{1 \text{ kg}}{1 \text{ g}}\right)\left(\frac{1 \text{ m}}{1 \text{ cm}}\right)^{-1}\left(\frac{1 \text{ s}}{1 \text{ s}}\right)^{-2} = (10^3)(10^2)^{-1} = 10.$$

So 1 Pascal = 10 CGS pressure.

Deducing Relations Among Physical Quantities

If we know which variables affect a physical quantity, we can sometimes deduce its form using dimensional analysis. For example, suppose the time period t of a simple pendulum depends on its length l, the mass of the bob m, and the acceleration due to gravity g. We assume a product relation of the form

$$t = k\, l^a\, m^b\, g^c, \tag{1}$$

where k is a dimensionless constant and a, b, c are exponents we want to evaluate. Taking dimensions on both sides:

$$T = L^a\, M^b\, (LT^{-2})^c = L^{a+c}\, M^b\, T^{-2c}.$$

Equating powers on both sides:

- For M: $b = 0$
- For T: $-2c = 1 \implies c = -\frac{1}{2}$
- For L: $a + c = 0 \implies a = \frac{1}{2}$

Substituting these values gives

$$t = k\sqrt{\frac{l}{g}}. \tag{2}$$

In the actual formula $k = 2\pi$, which is determined from the full physical analysis, not from dimensional analysis alone.

Limitations:

1. It only works for *product-type* dependencies. It cannot deduce equations where terms are added or subtracted (like $x = ut + \frac{1}{2}at^2$).
2. It cannot find numerical constants. The dimensional method left us with an unknown constant k in the pendulum equation. We need actual experiments (or deeper theory) to find that $k = 2\pi$.

CHAPTER 3

Motion

3.1 What is Motion?

Motion and Rest

The apparent change in a body's (or object's) position with time is called **motion**. If the body does not change its position as time passes, it is said to be at **rest**.

The relative nature of motion

But when do we say that a body is not changing its position? A cup on the table remains on the table — we say the cup is at rest. However, if we station ourselves on Mars, the whole Earth is found to be changing its position, and so the room, the table, and the cup are all continuously changing their positions.

- The cup is *at rest* if viewed from the room.
- The cup is *moving* if viewed from Mars.

Motion is therefore a combined property of the object under study and the observer. There is no meaning of rest or motion without the viewer.

Key Principle

Nothing is in absolute rest or in absolute motion.
Mars is moving with respect to the cup; the cup moves with respect to Mars.

Worked Example 3.1: Frames and relative motion

A passenger sits still inside a moving train.

(a) From the passenger's own frame: the passenger is *at rest*.

(b) From a bystander on the platform: the passenger is *moving* at the speed of the train.

(c) From an observer on the Moon: both the passenger and the bystander are moving (the Earth rotates and orbits the Sun).

All three descriptions are equally valid — rest and motion depend entirely on the reference frame of the observer.

Frame of reference

To locate the position of a particle we need a **frame of reference** — a choice of a spatial origin and coordinate axes to label position. We choose Cartesian coordinates, so the coordinates (x, y, z) of the particle P specify its position relative to that frame. A clock is added to the frame to measure time.

Rest and motion in terms of coordinates

Rest: If all three coordinates x, y, z of the particle remain unchanged as time passes, the particle is at rest with respect to this frame.

Motion: If any one or more coordinates change with time, the body is moving with respect to this frame.

One-dimensional motion first

For our first look at this subject we will forget about the three dimensions of the world and concentrate on moving in one direction — like a car on a single straight road. We will return to three dimensions after we see how to describe motion in one dimension.

3.2 Velocities and the Derivative

Position function

Suppose an object moves along a straight line. We describe its motion by an **equation of motion**

$$s = f(t),$$

where s is the displacement (direct distance from the origin along the line) of the object at time t. The function f that describes the motion is called the **position function** of the object.

Average velocity

In the time interval from t to $t + \Delta t$, the change in position (displacement) is $f(t + \Delta t) - f(t)$.

Average velocity

The average velocity over the time interval $[t,\, t + \Delta t]$ is

$$v_{\text{av}} = \frac{\text{displacement}}{\text{time}} = \frac{f(t + \Delta t) - f(t)}{\Delta t}.$$

Geometrically, this is the slope of the secant line PQ on the position-time graph.

Worked Example 3.2: Computing average velocity

A car moves along a straight road with position function $f(t) = 3t^2 + 2t$ metres (where t is in seconds).

(a) Displacement from $t = 1\,\text{s}$ to $t = 3\,\text{s}$:

$$f(1) = 3(1)^2 + 2(1) = 5 \text{ m}, \qquad f(3) = 3(3)^2 + 2(3) = 33 \text{ m}, \qquad \Delta s = 28 \text{ m}.$$

(b) Average velocity over $[1, 3]$:

$$v_{\text{av}} = \frac{f(3) - f(1)}{3 - 1} = \frac{28}{2} = 14 \text{ m/s}.$$

Try It Yourself 3.1: Average velocity from a table

A cyclist's position at various times is recorded below. Find the average velocity over the full 15 s interval.

t (s)	0	5	10	15
s (m)	0	12	20	24

(Answer: $v_{av} = 24/15 = 1.6$ m/s.)

Instantaneous velocity

Now suppose we compute the average velocities over shorter and shorter time intervals $[t, t + \Delta t]$ — in other words, we let Δt approach zero.

Instantaneous velocity

The instantaneous velocity $v(t)$ at time t is defined as the limit of these average velocities:

$$v(t) = \lim_{\Delta t \to 0} \frac{f(t + \Delta t) - f(t)}{\Delta t} = \frac{df}{dt},$$

provided that this limit exists. Geometrically, $v(t)$ equals the slope of the tangent line ℓ to the position-time curve at the point P.

Worked Example 3.3: Instantaneous velocity from the limit definition

Find the instantaneous velocity $v(t)$ of the car $f(t) = 3t^2 + 2t$ from the limit definition. Hence find $v(1)$ and $v(3)$.

Solution. Expand $f(t + \Delta t)$:

$$f(t + \Delta t) = 3(t + \Delta t)^2 + 2(t + \Delta t) = 3t^2 + 6t\,\Delta t + 3(\Delta t)^2 + 2t + 2\,\Delta t.$$

Subtract $f(t) = 3t^2 + 2t$:

$$f(t + \Delta t) - f(t) = 6t\,\Delta t + 3(\Delta t)^2 + 2\,\Delta t = \Delta t\,(6t + 3\,\Delta t + 2).$$

Divide by Δt and take $\Delta t \to 0$:

$$v(t) = \lim_{\Delta t \to 0} (6t + 3\,\Delta t + 2) = 6t + 2 \text{ m/s}.$$

At $t = 1\,\text{s}$: $v(1) = 6(1) + 2 = 8$ m/s. At $t = 3\,\text{s}$: $v(3) = 6(3) + 2 = 20$ m/s.

Note: the average velocity over $[1, 3]$ was $14\,\text{m/s}$, which lies between 8 and $20\,\text{m/s}$ — exactly as expected!

Try It Yourself 3.2: Limit definition for $f(t) = t^2$

Use the limit definition to show that for $f(t) = t^2$, the instantaneous velocity is $v(t) = 2t$ m/s. Hence find the velocity at $t = 3\,\text{s}$.

Hint: $(t + \Delta t)^2 - t^2 = 2t\,\Delta t + (\Delta t)^2$. *Answer:* $v(3) = 6$ m/s.

The derivative

The quantity df/dt found above is called the **derivative** of f with respect to t. The process of finding it is called **differentiating**.

Tangent line

The tangent line to $y = f(t)$ at the point $(t, f(t))$ is the unique line through $(t, f(t))$ whose slope equals $\dfrac{df(t)}{dt}$, the derivative of f at t.

Other notations for the derivative

Let $y = f(x)$. All of the following mean the same thing:

$$\frac{dy}{dx} = \frac{df}{dx} = \frac{d}{dx} f(x) = f'(x) = y'.$$

For derivatives with respect to time t, Newton used an overdot:

$$\dot{f} = \frac{df}{dt}, \qquad \dot{y} = \frac{dy}{dt}.$$

Notation	Name / origin
$\frac{dy}{dx}, \frac{df}{dx}, \frac{d}{dx}f(x)$	Leibniz notation
$f'(x),\ y'$	Lagrange (prime) notation
$\dot{y},\ \dot{f}$	Newton (dot) notation — time derivatives

All notations are in common use. In physics, $\dot{s} = v$ for velocity is popular for time derivatives. You should be comfortable reading all of them.

Formula Summary: Motion

Formula Summary: Motion

Quantity	Formula
Position function	$s = f(t)$
Displacement over $[t,\, t+\Delta t]$	$\Delta s = f(t+\Delta t) - f(t)$
Average velocity	$v_{\text{av}} = \dfrac{f(t+\Delta t) - f(t)}{\Delta t}$
Instantaneous velocity	$v(t) = \lim\limits_{\Delta t \to 0} \dfrac{f(t+\Delta t) - f(t)}{\Delta t} = \dfrac{df}{dt}$
Slope of tangent	$\dfrac{df}{dt}$ at that point

An important note on vectors: Because we are currently studying motion along a single straight line (1D), the direction of our vectors is indicated simply by a positive or negative sign. To keep the focus on the calculus, we will temporarily drop the vector notation ($\vec{v}$) and use scalars (v) to represent these 1D quantities.

3.3 Velocity as a Derivative and Graphing

Ex 1. A falling ball ($s = f(t) = 5t^2$).

t (s)	s (m)
0	0
1	5
2	20
3	45
4	80
5	125

The formula for this curve can be written as $s = f(t) = 5t^2$, which enables us to calculate the distance at any time.

If we want to know how fast a ball is falling at exactly $t = 4\,\text{s}$, we cannot simply use a wide interval. Between 3 and 4 seconds, the ball averages $35\,\text{m/s}$ ($v_{\text{av}} = \frac{80-45}{4-3}$). However, it is constantly accelerating during that time. That average does not reflect the true velocity at the exact 4-second mark.

To get closer to the truth, we look at a very small interval immediately following $t = 4\,\text{s}$:

- At $4.1\,\text{s}$: total displacement $s = 5(4.1)^2 = 84.05\,\text{m}$.

- At 4 s: total displacement was 80 m.
- Average velocity for this tiny window: $4.05/0.1 = 40.5\,\mathrm{m/s}$.

Taking the general approach, if at time $t_0 = 4\,\mathrm{s}$ the position is $s_0 = 5t_0^2$, then at $t_0 + \Delta t$ the new position is

$$s_0 + \Delta s = 5(t_0 + \Delta t)^2 = 5t_0^2 + 10t_0\,\Delta t + 5\,\Delta t^2.$$

So the extra displacement is $\Delta s = 10t_0\,\Delta t + 5\,\Delta t^2$, and the average velocity is

$$v_{\mathrm{av}} = \frac{\Delta s}{\Delta t} = 10t_0 + 5\,\Delta t. \tag{3}$$

Taking $\Delta t \to 0$:

$$v(\text{at time } t_0) = 10t_0.$$

At $t_0 = 4\,\mathrm{s}$: $v(4) = 10 \times 4 = 40\,\mathrm{m/s}$.

The procedure we have just carried out is performed so often in mathematics that special notation has been assigned. The velocity is

$$v = \lim_{\Delta t \to 0} \frac{\Delta s}{\Delta t} = \frac{ds}{dt}. \tag{4}$$

The quantity ds/dt is called the "derivative of s with respect to t". For our example:

$$\frac{d}{dt}(5t^2) = 10t.$$

A Short Table of Derivatives.

u, v, w are arbitrary functions of t; c and n are arbitrary constants.

Function, $f(t)$	**Derivative,** $\frac{df}{dt}$
c (constant)	0
t^n	$n\,t^{n-1}$
cu	$c\,\frac{du}{dt}$
$\sin(t)$	$\cos(t)$
$\cos(t)$	$-\sin(t)$
e^t	e^t
$\ln(t)$	$\frac{1}{t}$
$u + v + w + \cdots$	$\frac{du}{dt} + \frac{dv}{dt} + \frac{dw}{dt} + \cdots$
uv	$u\frac{dv}{dt} + v\frac{du}{dt}$
$\frac{u}{v}$	$\frac{v\,(du/dt) - u\,(dv/dt)}{v^2}$

Exercise. Find the derivative of the following functions.

a. $f(t) = 3t - 8$

b. $f(x) = mx + b$

c. $f(t) = 2.5t^2 + 6t$

d. $f(x) = 4 + 8x - 5x^2$

e. $A(p) = 4p^3 + \ln(p)$

f. $F(t) = e^t(t^2 - 1)$

g. $f(x) = \dfrac{1}{x^2 - 4}$

h. $F(v) = \dfrac{v}{v + 2}$

i. $g(u) = \dfrac{u + 1}{4u - 1}$

j. $f(x) = x^4 + \sin(x)$

k. $g(x) = \dfrac{1}{1 + \sqrt{x}}$

Example solutions:

$$\text{(a)} \quad f'(t) = 3, \qquad \text{(d)} \quad f'(x) = 8 - 10x.$$

3.4 Distance as an Integral

Now we discuss the inverse problem. Suppose that instead of a table of distances, we have a table of speeds at different times, starting from zero. From our earlier result, $s = 5t^2$ and $v = 10t$.

The distance travelled is equal to the area under the velocity–time curve. In general, when the velocity–time graph is not a simple geometric shape, we break the motion into small intervals Δt and approximate the distance in each interval as $\Delta s \approx v\,\Delta t$.

In general, suppose an object moves with velocity $v = v(t)$, where $a \le t \le b$. We take velocity readings at times $t_0(= a), t_1, t_2, \ldots, t_n(= b)$ so that the velocity is approximately constant on each sub-interval with $\Delta t = (b - a)/n$. The total distance is approximately

$$v(t_0)\Delta t + v(t_1)\Delta t + \cdots + v(t_{n-1})\Delta t = \sum_{i=0}^{n-1} v(t_i)\,\Delta t.$$

The true distance is the limit as $\Delta t \to 0$ (equivalently, $n \to \infty$):

$$s = \lim_{\Delta t \to 0} \sum_{i=0}^{n-1} v(t_i)\,\Delta t = \lim_{n \to \infty} \sum_{i=0}^{n-1} v(t_i)\,\Delta t. \tag{5}$$

The mathematicians have invented a symbol for this limit. The Δ turns into a d, the addition is written with a great "S" (from the Latin *summa*), distorted into the integral sign:

$$s = \int v(t)\,dt. \tag{6}$$

This process of adding all these terms together is called **integration**, and it is the opposite process to differentiation.

A Short Table of Derivatives and Integrals.

u, v, w are arbitrary functions of t; c, k, n are arbitrary constants.

Function, $f(t)$	**Derivative,** $\dfrac{df}{dt}$	**Integral,** $\int f(t)\,dt$
c	0	$ct + k$
t^n	$n\,t^{n-1}$	$\dfrac{t^{n+1}}{n+1} + k \quad (n \neq -1)$
cu	$c\,\dfrac{du}{dt}$	$c\int u\,dt$
$\sin(t)$	$\cos(t)$	$-\cos(t) + k$
$\cos(t)$	$-\sin(t)$	$\sin(t) + k$
e^t	e^t	$e^t + k$
$u + v + w + \cdots$	$\dfrac{du}{dt} + \dfrac{dv}{dt} + \cdots$	$\int u\,dt + \int v\,dt + \cdots$

Returning to our falling ball example, $v(t) = 10t$:

$$s = \int v(t)\,dt = \int 10t\,dt = 5t^2 + C.$$

The distance travelled between $t = 0$ and $t = 5\,\mathrm{s}$:

$$s = \int_0^5 v(t)\,dt = \int_0^5 10t\,dt = \left[5t^2\right]_0^5 = 5(5)^2 - 5(0)^2 = 125 \text{ m},$$

which agrees with the area under the velocity–time graph over $[0, 5]$.

Exercise. Evaluate the integral.

a. $\int (3t - 8)\,dt$

b. $\int (mx + b)\,dx$

c. $\int_0^3 (5t^2 + 6t)\,dt$

d. $\displaystyle\int_{\pi}^{4} (4 + 8x - 5x^2)\,dx$

e. $\displaystyle\int_{1}^{2} \big(3e^t + (t^2 - 1)\big)\,dt$

f. $\displaystyle\int_{0}^{\pi} (x^4 + \sin x)\,dx$

Example solutions:

(a) $\displaystyle\int (3t-8)\,dt = \tfrac{3}{2}t^2 - 8t + k,$ (c) $\displaystyle\int_0^3 (5t^2+6t)\,dt = \frac{5}{3}(3^3-0^3)+3(3^2-0^2) = 45+27 = 72.$

3.5 Acceleration

If the velocity of a particle remains constant as time passes, we say that it is moving with **uniform velocity**. If the velocity changes with time, the particle is said to be **accelerating**.

The **average acceleration** over $[t_1, t_2]$ is

$$a_{\text{av}} = \frac{v_2 - v_1}{t_2 - t_1} = \frac{\Delta v}{\Delta t}. \tag{7}$$

The **instantaneous acceleration** is

$$a(t) = \lim_{\Delta t \to 0} \frac{\Delta v}{\Delta t} = \frac{dv}{dt}. \tag{8}$$

Since velocity is ds/dt, and acceleration is the time derivative of velocity:

$$a(t) = \frac{dv}{dt} = \frac{d}{dt}\left(\frac{ds}{dt}\right) = \frac{d^2 s}{dt^2}, \tag{9}$$

which is the **second derivative** of position with respect to time.

For the falling ball $s(t) = 5t^2$:

$$v(t) = \frac{ds}{dt} = 10t, \qquad a(t) = \frac{dv}{dt} = 10 \text{ m/s}^2.$$

The acceleration is constant because the gravitational force on the falling body is constant, and by Newton's second law, acceleration is proportional to the net force.

3.6 Motion with Constant Acceleration

Suppose a particle moves along a straight line (taken as the x-axis) with constant acceleration a. Let the velocity at $t = 0$ be v_0 and at time t be v. Then:

$$\frac{dv}{dt} = a \implies dv = a\,dt.$$

Integrating from 0 to t (velocity from v_0 to v):

$$\int_{v_0}^{v} dv = \int_0^t a\,dt \implies v - v_0 = at,$$

$$\boxed{v = v_0 + at.} \tag{10}$$

Writing $dx/dt = v_0 + at$ and integrating from x_0 to x (time from 0 to t):

$$x - x_0 = v_0 t + \frac{1}{2}at^2,$$

$$\boxed{\Delta x = v_0 t + \tfrac{1}{2}at^2.} \tag{11}$$

Squaring equation (10): $v^2 = (v_0 + at)^2 = v_0^2 + 2av_0 t + a^2t^2 = v_0^2 + 2a\big(v_0 t + \frac{1}{2}at^2\big)$, so

$$\boxed{v^2 = v_0^2 + 2a\,\Delta x.} \tag{12}$$

Equations of Kinematics

$$v = v_0 + at, \qquad \Delta x = v_0 t + \tfrac{1}{2}at^2, \qquad v^2 = v_0^2 + 2a\,\Delta x.$$

The three equations above are called the **kinematic equations**. Remember that x represents the position (or displacement) at time t and not necessarily the total distance travelled. The quantities v_0, v, and a may take positive or negative values depending on their direction.

3.7 Freely Falling Bodies

A common example of motion in a straight line with constant acceleration is the **free fall** of a body near the earth's surface. If air resistance is neglected and a body is dropped, it falls along a vertical straight line. The acceleration in the vertically downward direction has magnitude approximately $9.8\,\mathrm{m/s^2}$ and is known as the **acceleration due to gravity**, denoted by g.

The sign of g depends on the chosen positive direction. Taking vertically upward as positive gives $a = -g$; taking downward as positive gives $a = +g$. Substituting into the kinematic equations accordingly:

$$v = v_0 \pm g\,t, \qquad \Delta y = v_0 t \pm \tfrac{1}{2}g\,t^2, \qquad v^2 = v_0^2 \pm 2g\,\Delta y,$$

where the upper sign $(+)$ applies when downward is taken as positive, and the lower sign $(-)$ when upward is taken as positive.

Motion in a Plane (2D) and in Space (3D). In three dimensions the position vector is

$$\vec{s} = x\hat{x} + y\hat{y} + z\hat{z}.$$

Differentiating with respect to time gives the velocity vector:

$$\vec{v} = \frac{d\vec{s}}{dt} = \frac{dx}{dt}\hat{x} + \frac{dy}{dt}\hat{y} + \frac{dz}{dt}\hat{z} = v_x\hat{x} + v_y\hat{y} + v_z\hat{z},$$

with magnitude (speed)

$$v = |\vec{v}| = \sqrt{v_x^2 + v_y^2 + v_z^2}.$$

Differentiating once more gives the acceleration vector:

$$\vec{a} = \frac{d\vec{v}}{dt} = a_x\hat{x} + a_y\hat{y} + a_z\hat{z}, \qquad a = \sqrt{a_x^2 + a_y^2 + a_z^2}.$$

3.8 Projectile Motion

An important example of motion in a plane with constant acceleration is **projectile motion**. When a particle is thrown obliquely near the earth's surface, it moves along a curved path. Such a particle is called a **projectile**. We assume the particle remains close to the surface and air resistance is negligible, so acceleration is constant in the vertically downward direction with magnitude $g \approx 9.8\,\mathrm{m/s^2}$.

A particle is projected from point O with initial speed v_0 at angle θ with the horizontal. The initial conditions are:

$$x_0 = 0, \quad v_{0,x} = v_0\cos\theta, \quad a_x = 0, \tag{13}$$

$$y_0 = 0, \quad v_{0,y} = v_0\sin\theta, \quad a_y = -g. \tag{14}$$

Horizontal Motion ($a_x = 0$):

$$v_x = v_0\cos\theta \qquad \text{(constant)}, \tag{15}$$

$$x = v_0 t\cos\theta. \tag{16}$$

Vertical Motion:

$$v_y = v_0\sin\theta - g\,t, \tag{17}$$

$$y = v_0 t\sin\theta - \tfrac{1}{2}g\,t^2, \tag{18}$$

$$v_y^2 = (v_0\sin\theta)^2 - 2g\,y. \tag{19}$$

Time of Flight. At point B (landing), $y = 0$:

$$0 = v_0 t\sin\theta - \tfrac{1}{2}g\,t^2 = t\big(v_0\sin\theta - \tfrac{1}{2}g\,t\big).$$

The non-trivial solution gives the time of flight:

$$T = \frac{2v_0\sin\theta}{g}. \tag{20}$$

Range. The horizontal range OB is the distance travelled during time T:

$$OB = v_0\,T\cos\theta = \frac{2v_0^2\sin\theta\cos\theta}{g} = \frac{v_0^2\sin 2\theta}{g}. \tag{21}$$

Notice that the greatest range is achieved if the projectile is thrown at $45°$ to the horizontal.

Maximum Height Reached. At the peak, $v_y = 0$. From (18): $t = v_0 \sin\theta / g$. Substituting into (19):

$$H = \frac{v_0^2 \sin^2\theta}{2g}. \tag{22}$$

The time to reach maximum height is exactly half the total time of flight.

Equation of Trajectory. Eliminating t between (16) and (19):

$$y = x \tan\theta - \frac{g\,x^2}{2v_0^2 \cos^2\theta}. \tag{23}$$

This is the equation of the trajectory — a **parabola**.

CHAPTER **4**

Newton's Three Laws of Motion

Before stating any law, it is worth asking a simple question. What does it take for something to keep moving? If you roll a ball across the floor, what eventually stops it?

The instinctive answer, and the one given by Aristotle for well over a thousand years, is that motion requires a continuous sustaining force. Remove the force and the motion should stop. This view matches what we see every day: carts roll to a halt, thrown balls fall back, skidding cars come to rest. The trouble is that it is wrong. What actually stops a rolling ball is friction, not the absence of some hidden push. In the absence of friction (or any other opposing force), the ball would continue rolling forever. This is the content of Newton's first law, but it took from Aristotle to Galileo to articulate it clearly, and from Galileo to Newton to place it in its proper setting.

4.1 The First Law (the Law of Inertia)

The First Law

In the absence of a net external force, a body at rest remains at rest, and a body in motion continues to move at constant velocity.

The tendency of a body to continue in its initial state of motion is called its **inertia**. Accordingly, the First Law is often called the **Law of Inertia**. Written as an equation:

$$\sum \vec{F} = 0 \iff \vec{a} = 0. \tag{24}$$

This is less a statement about motion than a statement about force: force is not what *maintains* motion, but what *changes* it. A body travelling at constant velocity is no more mysterious than a body at rest — both are states of zero acceleration, and both require zero net force.

There is a subtlety in the first law that is easy to miss. The first law holds only in an **inertial frame** — a frame that is not itself accelerating. The car that brakes suddenly is not such a frame; viewed from the ground, the ball on the dashboard is simply continuing in a straight line at constant velocity while the car decelerates around it.

Is the Earth an inertial frame? Strictly speaking, no. But quantitatively: the Earth's orbital radius is $R \approx 1.5 \times 10^{11}$ m and its orbital period is $T \approx 3.15 \times 10^{7}$ s, giving an orbital speed

$$v = \frac{2\pi R}{T} \approx 2.99 \times 10^{4}\ \mathrm{m/s} \tag{25}$$

and a centripetal acceleration

$$a = \frac{v^2}{R} \approx 6 \times 10^{-3}\ \mathrm{m/s^2}. \tag{26}$$

Compared to $g = 9.8\,\mathrm{m/s^2}$, this is smaller by a factor of roughly 1600. Newton's first law therefore holds as an extraordinarily good approximation, and we will use it as if it were exact.

4.2 The Second Law

The Second Law

For any particle of mass m, the net force $\vec{F}$ on the particle is always equal to the mass m times the particle's acceleration:

$$\vec{F} = m\vec{a}. \tag{27}$$

Three points are worth emphasising:

1. $\vec{F}$ is the *net* force — the vector sum of *all* forces acting on the body.

2. The equation is a *vector equation*: it must hold component by component, and in particular the acceleration vector always points in the same direction as the net force.

3. Despite appearances, this is *not* a definition of force. Force has an independent physical meaning, and the second law is the (enormously successful) empirical claim that force and acceleration are proportional, with the constant of proportionality being the body's mass.

As a simple application, consider two blocks of mass m_1 and m_2 connected by a light inextensible rope on a frictionless horizontal surface, with a horizontal force F applied to m_2 pulling the pair along. Treating the pair as a single system of mass $M = m_1 + m_2$, the acceleration is

$$a = \frac{F}{m_1 + m_2}. \tag{28}$$

To find the tension in the rope we isolate m_1. The only horizontal force on m_1 is the tension T, so

$$T = m_1\, a = \frac{m_1 F}{m_1 + m_2}. \tag{29}$$

This example illustrates a general strategy. When several bodies are connected, it often pays to treat them first as a single system (to find the common acceleration), and then isolate each body in turn (to find internal forces like tension).

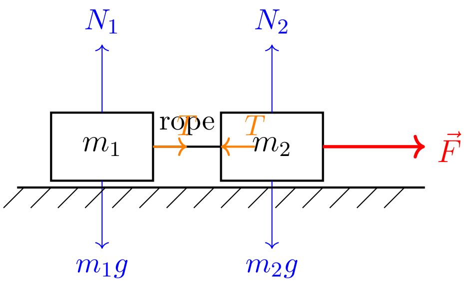


Figure 4.1: Two blocks connected by a light rope on a frictionless surface. Horizontally, m_1 is pulled only by the tension T, while m_2 is pulled forward by F and backward by T.

4.3 The Third Law

The Third Law

If object 1 exerts a force $\vec{F}_{21}$ on object 2, then object 2 will exert a reaction force $\vec{F}_{12}$ on object 1 given by

$$\vec{F}_{12} = -\vec{F}_{21}. \tag{30}$$

The crucial point, and the one most often misunderstood, is that these two forces act on **different bodies**. They therefore never cancel each other.

Let us apply this to a familiar situation. You drop an apple. The Earth pulls the apple downward with gravitational force $F = mg$. By the third law, the apple must pull the Earth upward with an equal force. The apple's acceleration is

$$a_{\text{apple}} = \frac{F}{m_{\text{apple}}} = g \approx 9.8 \text{ m/s}^2. \tag{31}$$

The Earth's acceleration is

$$a_{\text{Earth}} = \frac{F}{M_{\text{Earth}}} = \frac{mg}{M_{\text{Earth}}}. \tag{32}$$

Taking a 100 g apple and $M_{\text{Earth}} \approx 5.97 \times 10^{24}$ kg:

$$a_{\text{Earth}} \approx 1.6 \times 10^{-25} \text{ m/s}^2. \tag{33}$$

Over the roughly half-second duration of the apple's fall, the Earth moves toward the apple by a distance far smaller than the diameter of a proton.

The principle scales up. The third law is also at the root of the **horse-cart paradox**. A horse pulls a cart forward. By the third law, the cart pulls the horse backward with an equal force. So how does anything ever move? The resolution: the forward pull on the cart and the backward pull on the horse act on *different bodies*. To understand the horse's motion we must look at all forces on the horse, which includes not only the backward pull from the cart but also the forward friction from the ground on its hooves. Both the horse and cart accelerate together. No paradox, only careful bookkeeping.

Class Exercise: The Atwood Machine

Two blocks are connected by a light, inextensible string passing over a small frictionless pulley fixed to the ceiling. Block 1 has mass $m_1 = 6\,\text{kg}$ and Block 2 has mass $m_2 = 4\,\text{kg}$. The system is released from rest. Take $g = 10\,\text{m/s}^2$.

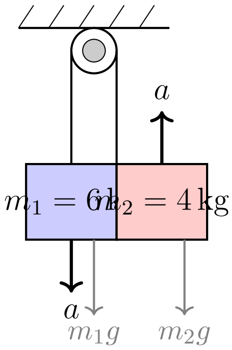


Figure 4.2: The Atwood Machine. Block 1 ($m_1 = 6\,\text{kg}$) descends while Block 2 ($m_2 = 4\,\text{kg}$) ascends.

(a) Draw a free body diagram for each block and write down Newton's second law for each.

(b) Hence find the acceleration a of the system and the tension T in the string.

(c) What is the tension in the string if both masses were equal? Comment on your answer physically.

CHAPTER 5

Further Applications of Newton's Laws

5.1 Contact Forces

Pushing, lifting and pulling are ***contact forces*** that we experience in the everyday world. Rest your hand on a table; the atoms that form the molecules that make up the table and your hand are in contact with each other. If you press harder, the atoms are also pressed closer together. The electrons in the atoms begin to repel each other and your hand is pushed in the opposite direction by the table.

According to Newton's Third Law, the force of your hand on the table is equal in magnitude and opposite in direction to the force of the table on your hand. Clearly, if you push harder the force increases. If you push your hand straight down on the table, the table pushes back in a direction perpendicular (normal) to the surface. Slide your hand gently forward along the surface of the table. You barely feel the table pushing upward, but you do feel the friction acting as a resistive force to the motion of your hand. This force acts tangential to the surface and opposite to the motion of your hand. Push downward and forward. Try to estimate the magnitude of the force acting on your hand.

The force of the table acting on your hand, $\vec{F}_c \equiv \vec{C}$, is called the ***contact force***. This force has both a normal component to the surface, $\vec{C}_\perp \equiv \vec{\mathscr{N}}$, called the ***normal force***, and a tangential component to the surface, $\vec{C}_\parallel \equiv \vec{f}$, called the ***friction force*** (Figure 5.1).

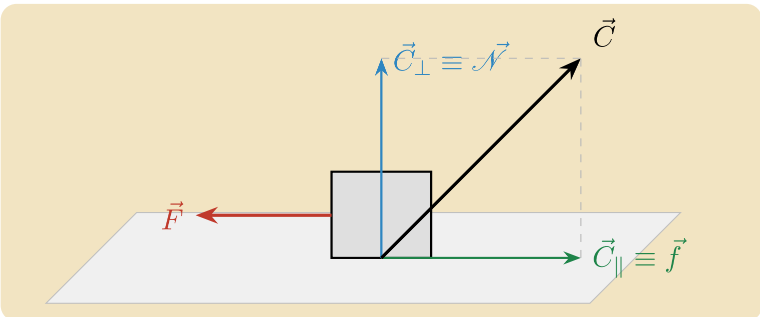


Figure 5.1: The block is pushed to the left by $\vec{F}$. The contact force $\vec{C}$ decomposes into a normal component $\vec{\mathscr{N}}$ and a tangential friction component $\vec{f}$ opposing the motion.

The contact force, written in terms of its component forces, is therefore

$$\vec{C} = \vec{C}_\perp + \vec{C}_\parallel \equiv \vec{\mathscr{N}} + \vec{f}\,. \tag{34}$$

The normal component $\vec{\mathscr{N}}$ and the tangential component $\vec{f}$ are not independent forces but the vector components of the contact force, perpendicular and parallel to the surface of contact.

Friction

When a block slides along a horizontal surface or down an inclined plane, a lateral force resists the motion. Even when the block is at rest on an inclined plane, a lateral force prevents it from sliding. This resistive force is known as *dry friction*, and there are two distinguishing types when surfaces are in contact with each other.

When two surfaces move relative to one another, the friction is called ***kinetic friction*** (or ***sliding friction***). When the surfaces are stationary relative to each other but a lateral force is present, the friction is called ***static friction***.

Leonardo da Vinci was the first to record systematic measurements of kinetic friction, over a period spanning 1493 to about 1515. His results were rediscovered and published by Guillaume Amontons in 1699, and extended by Charles-Augustin Coulomb, who showed that kinetic friction is also independent of sliding speed for ordinary speeds.

Kinetic Friction

The magnitude of the kinetic friction force $\vec{f}^k$ is proportional to the normal force between the two surfaces,

$$f_k = \mu_k \mathscr{N} \,, \tag{35}$$

where μ_k is the ***coefficient of kinetic friction***, a dimensionless constant depending only on the two materials in contact. A notable result is that f_k is independent of the apparent contact area. Two blocks of the same mass but different footprints require the same force to slide at constant speed (Figure 5.2). This follows from the microscopic structure of surfaces, as discussed in the next section.

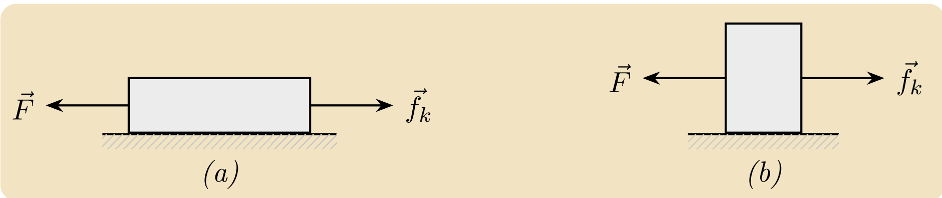


Figure 5.2: Kinetic friction is independent of contact area. Block (a) has twice the footprint of block (b), yet both require the same force to slide at constant speed.

Static Friction

Static friction does not take a fixed value; it adjusts itself to prevent relative motion. As you push a stationary block with increasing force, static friction matches your push exactly until a maximum value is reached. Beyond this threshold, the block slips and kinetic friction takes over. The magnitude of static friction therefore satisfies

$$0 \le f_s \le (f_s)_{\max} \,, \tag{36}$$

where the maximum is found experimentally to be proportional to the normal force,

$$(f_s)_{\max} = \mu_s \mathscr{N} \,. \tag{37}$$

Here μ_s is the ***coefficient of static friction***. Experiment consistently gives $\mu_s > \mu_k$: more force is needed to initiate sliding than to sustain it. This small difference accounts for the stick-slip behaviour of chalk on a blackboard, fingernails on glass, or a violin bow on a string.

The direction of static friction always opposes the applied force, as shown in Figure 5.3.

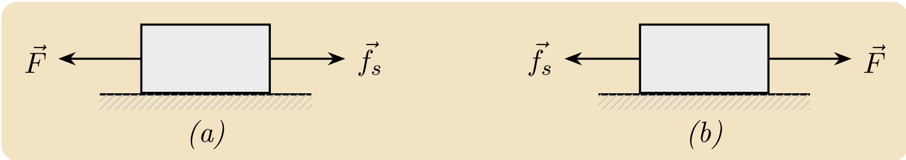


Figure 5.3: Static friction opposes the applied force. In (a) $\vec{F}$ acts to the left so $\vec{f}_s$ acts to the right. In (b) the roles are reversed.

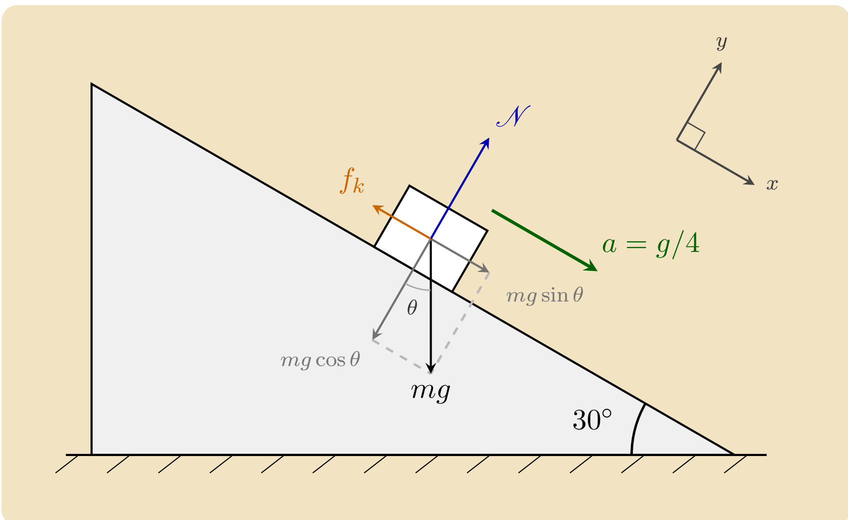
Two important distinctions between static and kinetic friction are worth keeping in mind.

1. The magnitude and direction of static friction depend on the applied forces and adjust to maintain equilibrium, up to the maximum $(f_s)_{\max}$. Kinetic friction, by contrast, has a fixed magnitude $\mu_k\mathcal{N}$ for a given normal force.
2. When the applied force along the contact surface reaches $(f_s)_{\max}$, the surfaces begin to slip. This threshold condition is called the ***just slipping*** condition, and marks the transition from static to kinetic friction.

Worked Example 5.1: Sliding Block

A block of mass m slides down an incline of angle $30°$ with an acceleration $g/4$. Find the kinetic friction coeffcient μ_k.

Solution. We resolve the forces into components parallel and perpendicular to the incline, taking the positive direction down the slope for motion and outward from the surface for the normal direction.



Perpendicular to the incline there is no acceleration, so the normal force balances the perpendicular component of gravity,

$$\mathscr{N} - mg\cos\theta = 0 \implies \mathscr{N} = mg\cos\theta.$$

Along the incline, the block accelerates down the slope at $a = g/4$. The driving force is the parallel component of gravity, $mg\sin\theta$, opposed by kinetic friction $f_k = \mu_k \mathscr{N}$ acting up the slope. Newton's second law gives

$$mg\sin\theta - \mu_k\mathscr{N} = ma.$$

Substituting $\mathscr{N} = mg\cos\theta$ and $a = g/4$,

$$mg\sin\theta - \mu_k mg\cos\theta = \tfrac{1}{4}mg.$$

The mass m and the acceleration g cancel throughout, leaving a relation among pure numbers,

$$\sin\theta - \mu_k\cos\theta = \tfrac{1}{4}.$$

Solving for the coefficient of kinetic friction,

$$\mu_k = \frac{\sin\theta - \frac{1}{4}}{\cos\theta}.$$

With $\theta = 30°$, so that $\sin\theta = \frac{1}{2}$ and $\cos\theta = \frac{\sqrt{3}}{2}$,

$$\mu_k = \frac{1/2 - 1/4}{\sqrt{3}/2} = \frac{1}{2\sqrt{3}}.$$

The Normal Force

When two solid bodies are pressed together, their surfaces resist interpenetration. Push your hand against a wall and the wall pushes back, preventing your hand from passing through (Figure 5.4). This perpendicular push is called the *normal force*, where *normal* means perpendicular to the surface of contact. Notice that the harder you push on the wall with a force $\vec{F}$, the harder the wall pushes back on your hand with the normal force $\vec{\mathscr{N}}$.

The origin of this force is atomic. The atoms at the surface of your hand and the atoms at the surface of the wall repel one another the moment they come into contact, opposing any further compression. The material of the wall behaves, at the microscopic level, like an array of tiny springs.[1] When you push against the wall, these atomic springs compress slightly, and the repulsive force they exert grows with the compression until it exactly balances your push. For a hard material such as concrete, the compression is so small as to be invisible, and the wall appears perfectly rigid.

This is why the normal force is not a fixed quantity: it adjusts itself to match whatever force is pressing the surfaces together. Push harder, and the atomic springs compress more, increasing the normal force in response. As the figure shows, $\vec{F}$ and $\vec{\mathscr{N}}$ always grow together and remain equal in magnitude. The normal force is therefore a *constraint force*, one whose magnitude is determined not by a fixed law but by the condition that the two surfaces do not pass through each other.

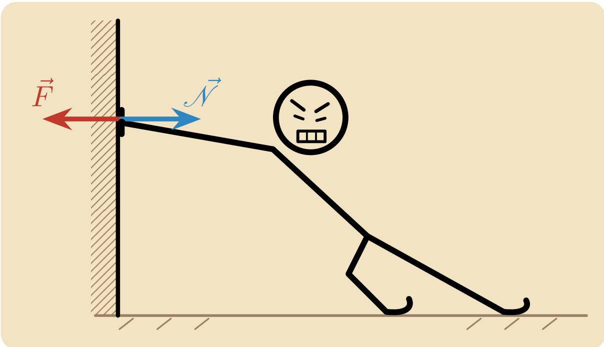


Figure 5.4: When you push against a wall ($\vec{F}$), the wall pushes back on your hand with an equal and opposite normal force ($\vec{\mathscr{N}}$), perpendicular to its surface. Both forces act at the point of contact.

The Normal Force and Weight

We say we are measuring our weight when we step onto a bathroom scale. But what is the scale actually measuring?

Two forces act on a person standing on a scale (Figure 5.1.2). The first is the gravitational force $\vec{F}_g = m\vec{g}$, directed downward. The second is the contact force from the scale;

[1]The "atomic springs" picture is a useful analogy but not the complete story. The true origin of this repulsion is quantum mechanical: as electron clouds overlap, the Pauli exclusion principle forces electrons into higher energy states, giving rise to a repulsive force. Classical electrostatics plays a secondary role. We will encounter the Pauli exclusion principle when we study modern physics.

because the person is at rest, this contact force is purely normal and we write it as $\vec{\mathscr{N}}$, directed upward. Since the person is at rest, the vertical acceleration is zero, and Newton's Second Law gives

$$\vec{\mathscr{N}} + \vec{F}_g = \vec{0}. \tag{38}$$

Taking the positive direction upward, this becomes

$$\mathscr{N} - mg = 0 \implies \mathscr{N} = mg. \tag{39}$$

It is worth being precise about four quantities that are easy to confuse. The *true mass* m is the amount of matter in the person, an invariant property that does not change with motion or location.[2] The *true weight* mg is the actual gravitational force the Earth exerts on the person. The scale, however, does not measure either of these directly. It measures the normal force $\mathscr{N}$ pressing on it, which we call the *apparent weight*. Dividing by g, it displays a number in kilograms, the *apparent mass* $\mathscr{N}/g$. At rest the apparent mass equals the true mass and the distinction seems academic. The elevator, discussed in the next section, will reveal the difference.

The word "weight" is often used loosely to mean the gravitational force the Earth exerts on an object. We shall always call that force the *gravitational force* rather than "weight", to keep it distinct from what a scale reads. When you jump into the air you feel "weightless" because no normal force acts on you, even though the Earth still pulls you down. You have not turned gravity off.

A common misconception deserves to be addressed before we proceed. The normal force and the gravitational force are two completely different forces. They are equal in magnitude here only because the person is at rest, and this equality will fail the moment there is acceleration. More importantly, they do not form a Newton's Third Law interaction pair: both act on the same object (the person), whereas Third Law pairs always act on *different* objects. The true Third Law partner of the gravitational force on the person due to the Earth, $\vec{F}^g_{E,P}$, is the gravitational force on the Earth due to the person, $\vec{F}^g_{P,E}$, and they satisfy $\vec{F}^g_{E,P} = -\vec{F}^g_{P,E}$. These act on different objects: one on the person, the other on the Earth. Likewise, the Third Law partner of the normal force $\vec{\mathscr{N}}$ from the scale is the force the person's feet exert downward on the scale surface.

Experience in an Elevator

Consider a person of mass $m = 60\,\text{kg}$ standing on a scale inside an elevator. In each case exactly two forces act on the person: the gravitational force $m\vec{g}$ directed downward and the normal force $\vec{\mathscr{N}}$ directed upward.

(a) At Rest or Moving at Constant Velocity.

[2] Newtonian mechanics treats mass as a fixed quantity. Special relativity shows that an object's inertia grows with its speed. One common way to express this introduces a velocity-dependent mass $m = m_0/\sqrt{1 - v^2/c^2}$, where m_0 is the rest mass and c is the speed of light; for everyday speeds $v \ll c$ this is indistinguishable from m_0. Modern physics usually prefers instead to keep mass as the invariant m_0 and describe the growing inertia through momentum and energy. Either way, the effect is utterly negligible at ordinary speeds. We will return to this when we study modern physics.

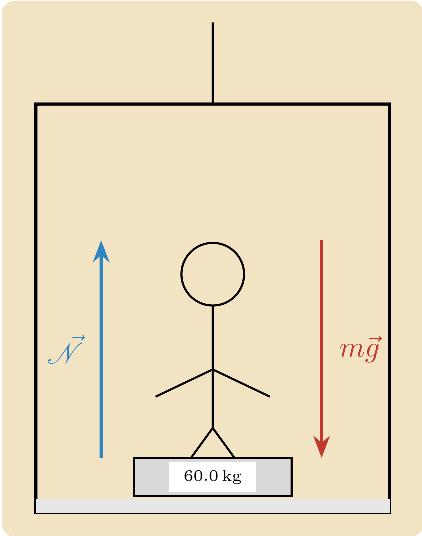
When the elevator is stationary or moving at constant velocity, the acceleration is zero. The person's state of motion is not changing, so the floor need only support the person's weight, nothing more. Newton's second law in the vertical direction gives

$$\vec{\mathscr{N}} + \vec{F}_g = \vec{0} \implies \mathscr{N} - mg = 0,$$

so that

$$\mathscr{N} = mg = (60)(10) = 600 \text{ N}.$$

The scale displays $\mathscr{N}/g = 60.0\,\text{kg}$. Here the apparent weight equals the true weight, which is why we never notice the distinction in ordinary standing.



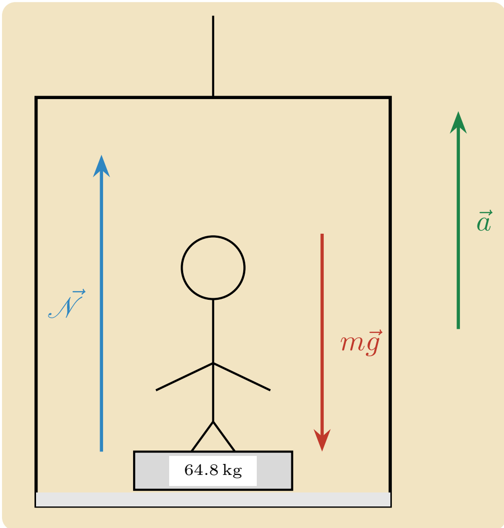
(b) Accelerating Upward.

Now the elevator accelerates upward. To make the person accelerate upward too, there must be a net upward force on them. But the person's inertia resists this change in motion: their body "wants" to stay at rest or moving as it was. The floor must therefore push up harder than before, hard enough both to support the weight and to supply the extra upward force the acceleration demands. The normal force rises above mg. Newton's second law gives

$$\mathscr{N} - mg = ma \implies \mathscr{N} = m(g + a). \qquad (40)$$

Suppose the scale reads 64.8 kg, meaning $\mathscr{N} = 648\,\text{N}$:

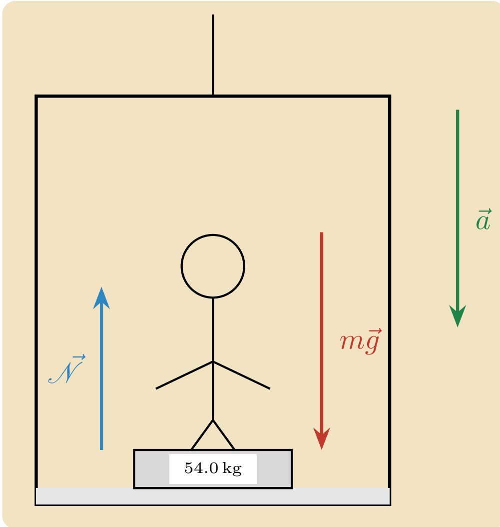
$$648 = 60(10 + a) \implies a = 0.8 \text{ m s}^{-2} \text{ upward}.$$



The person feels heavier because the floor genuinely presses harder on their feet. The true weight is unchanged at 600 N; only the apparent weight has risen, to 648 N.

(c) Accelerating Downward.

When the elevator accelerates downward, the person's inertia resists the change: the body "wants" to keep moving as it was, so for a moment it does not fall away with the floor as readily as the floor drops beneath it. The person presses less firmly on the scale, and so the scale presses less firmly back. The normal force falls below mg, leaving a net downward force that produces the downward acceleration:



$$mg - \mathscr{N} = ma \implies \mathscr{N} = m(g - a). \quad (41)$$

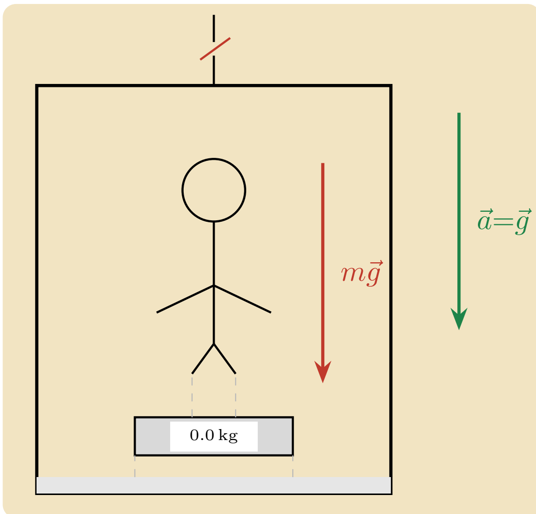
If the scale reads 54.0 kg, meaning $\mathscr{N} = 540\,\mathrm{N}$:

$$540 = 60(10 - a) \implies a = 1.0\,\mathrm{m\,s^{-2}} \text{ downward.}$$

The person feels lighter because the floor really does press less on their feet. The true weight is unchanged at 600 N; only the apparent weight has fallen, to 540 N.

(d) Free Fall.

Now imagine the cable snaps and the elevator falls freely. Gravity acts on everything inside equally, so the person, the scale, and the floor all accelerate downward together at the same rate $a = g$. Because the floor falls away just as fast as the person does, it never has to push on them at all. The normal force drops all the way to zero. Setting $a = g$ in equation (41),



$$\mathscr{N} = m(g - g) = 0.$$

The scale reads zero, yet gravity has certainly not switched off; the Earth still pulls the person down with the full force mg. The person simply floats inside the falling car because there is no longer any contact force to register. This is exactly what we mean by *weightlessness*: not the absence of gravity, but the absence of the normal force.

A Glimpse of Something Deeper (Supplementary)

We close with a question that troubled Einstein and ultimately led him to general relativity. Imagine you wake up inside a sealed box with no windows. You feel a normal force pressing up on your feet. How could you tell whether you are sitting stationary on the surface of the Earth, with gravity pulling you down, or drifting in empty space far from any planet, with the box accelerating upward at $g = 9.8\,\mathrm{m/s}^2$?

The remarkable answer is that you cannot. No mechanical experiment conducted inside the box will distinguish between these two situations. Drop a ball: in both cases it accelerates toward the floor at $10\,\mathrm{m/s}^2$. Step on a scale: it reads mg in both cases. Einstein elevated this to a fundamental principle: **gravity and accelerated motion are locally equivalent**. This is the **equivalence principle**.

a) Two Kinds of Mass. The symbol m appears in two logically distinct roles. In Newton's second law, m appears as **inertial mass**: the resistance of an object to acceleration. In the gravitational force law, m appears as **gravitational mass**: the quantity that determines how strongly gravity pulls on the object. There is no obvious reason these two quantities should be equal. Yet every experiment ever performed finds them equal. The Hungarian physicist Loránd Eötvös demonstrated this equality to one part in 10^8 in the 1880s, and modern experiments push this to one part in 10^{15}.

The equality $m_{\text{inertial}} = m_{\text{gravitational}}$ is exactly what ensures that all objects fall at the same rate regardless of their mass.

b) Light Must Bend. The equivalence principle has a striking consequence that goes beyond mechanics. Shine a horizontal beam of light across the accelerating box. While the light travels across, the box accelerates upward, so the beam hits the far wall slightly below where it was aimed — the light beam appears to curve downward. Since an accelerating box is locally indistinguishable from a gravitational field, gravity must also bend light. This was a radical prediction: Newtonian gravity acts only on objects with mass, and light was thought to be massless. Einstein's prediction was confirmed experimentally in 1919 by Arthur Eddington, who photographed stars near the edge of the Sun during a solar eclipse and found that their apparent positions were shifted by exactly the amount Einstein predicted. The announcement made Einstein famous overnight.

c) Where This Leads. Once you take the equivalence principle seriously and ask what geometry of space and time is consistent with it, you are led to **general relativity**. Gravity is reinterpreted not as a force but as the curvature of spacetime caused by mass and energy. Objects in free fall follow the straightest possible paths through this curved geometry. The bending of light, the slowing of clocks in gravitational fields, the existence of gravitational waves, and the physics of black holes all follow from the same geometric framework. The elevator we analysed today is the first step in that journey.

5.2 Circular Motion

We shall now investigate a special class of motions: motion in a plane about a central point, which we refer to as *central motion*. The most important special case is circular motion. There are many instances of circular motion: a bicycle rider on a circular track, a ball spun around on a string, and the rotation of a spinning wheel are just a few examples. The motion of the Moon around the Earth is nearly circular. The motions of the planets around the Sun are nearly circular. Our Sun moves in a nearly circular orbit about the centre of our galaxy.

We begin by describing the kinematics of circular motion – the position, velocity, and acceleration – as a special case of two-dimensional motion. Unlike linear motion, where velocity and acceleration are directed along the line of motion, in circular motion the velocity is always tangent to the circle. This means that as the object moves in a circle, the direction of the velocity is always changing. Examining this change reveals that the direction of the acceleration is towards the centre of the circle. This inward component of acceleration is called the *centripetal acceleration*. If the object is also changing speed, there is an additional non-zero tangential acceleration in the direction of motion. For uniform circular motion (constant speed), only the centripetal acceleration is non-zero.

Newton recognised that the Moon is always "falling" towards the Earth; otherwise, by the First Law, it would continue in a straight line rather than follow a circular orbit. There must therefore exist a centripetal force, a radial force pointing inward, producing the centripetal acceleration. Because Newton's Second Law is a vector equation, the radial component gives

$$F_r = ma_r. \tag{42}$$

Circular Motion: Velocity and Angular Velocity

We describe circular motion using polar coordinates. The position vector $\vec{r}(t)$ of an object moving in a circular orbit of radius r is written as

$$\vec{r}(t) = r\,\hat{r}(t), \tag{43}$$

where $\hat{r}(t)$ is the radial unit vector pointing from the origin to the object. At any point P with coordinates $(r, \theta(t))$, we define two sets of unit vectors: the polar basis $(\hat{r}(t),\, \hat{\theta}(t))$ and the Cartesian basis $(\hat{\imath},\, \hat{\jmath})$, as shown in Figure 5.5. Their relationship is

$$\hat{r}(t) = \cos\theta(t)\,\hat{\imath} + \sin\theta(t)\,\hat{\jmath}, \tag{44}$$

$$\hat{\theta}(t) = -\sin\theta(t)\,\hat{\imath} + \cos\theta(t)\,\hat{\jmath}. \tag{45}$$

Before computing the velocity, we calculate the time derivatives of the unit vectors. Using the chain rule on Eq. (44),

$$\frac{d\hat{r}(t)}{dt} = (-\sin\theta(t)\,\hat{\imath} + \cos\theta(t)\,\hat{\jmath})\,\frac{d\theta}{dt} = \frac{d\theta}{dt}\,\hat{\theta}(t). \tag{46}$$

Similarly, differentiating Eq. (45),

$$\frac{d\hat{\theta}(t)}{dt} = (-\cos\theta(t)\,\hat{\imath} - \sin\theta(t)\,\hat{\jmath})\,\frac{d\theta}{dt} = -\frac{d\theta}{dt}\,\hat{r}(t). \tag{47}$$

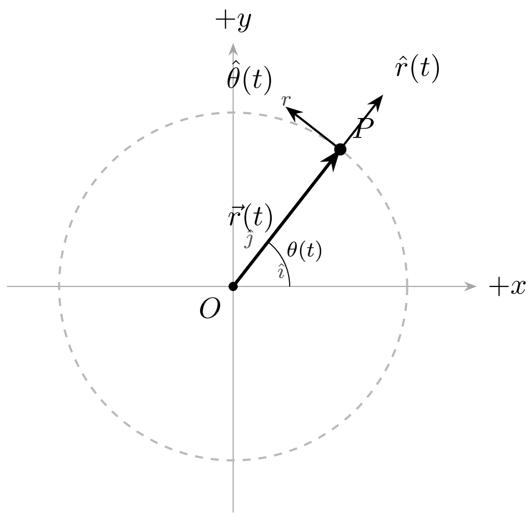


Figure 5.5: A circular orbit of radius r with the polar unit vectors $\hat{r}(t)$ and $\hat{\theta}(t)$ at the point P.

The velocity vector is then

$$\vec{v}(t) = \frac{d\vec{r}}{dt} = r\frac{d\hat{r}}{dt} = r\frac{d\theta}{dt}\,\hat{\theta}(t) = v_\theta\,\hat{\theta}(t), \tag{48}$$

where the tangential component of the velocity is

$$v_\theta = r\frac{d\theta}{dt}. \tag{49}$$

The *angular speed* is defined as the magnitude of the rate of change of angle,

$$\omega \equiv \left|\frac{d\theta}{dt}\right|, \tag{50}$$

so that the speed is $v = |\vec{v}| = r\omega$.

Geometric Derivation of the Velocity

Consider a particle undergoing circular motion. During a time interval Δt, the particle moves from position $\vec{r}(t)$ to $\vec{r}(t + \Delta t)$, sweeping an angle $\Delta\theta$, as shown in Figure 5.6.

The magnitude of the displacement is $|\Delta\vec{r}| = 2r\sin(\Delta\theta/2)$. For small angles, $\sin(\Delta\theta/2) \approx \Delta\theta/2$, so

$$|\Delta\vec{r}| \approx r\,\Delta\theta. \tag{51}$$

This is consistent with the limit $\Delta\theta \to 0$, where the chord approaches the arc length $r\,\Delta\theta$. The magnitude of the velocity then follows as

$$v \equiv |\vec{v}(t)| = \lim_{\Delta t\to 0}\frac{|\Delta\vec{r}|}{\Delta t} = r\lim_{\Delta t\to 0}\frac{\Delta\theta}{\Delta t} = r\frac{d\theta}{dt} = r\omega. \tag{52}$$

As $\Delta t \to 0$, the direction of $\Delta\vec{r}$ approaches the tangent to the circle, perpendicular to $\vec{r}(t)$ and in the $+\hat{\theta}$-direction for counterclockwise motion. This confirms that the velocity is always tangent to the circle.

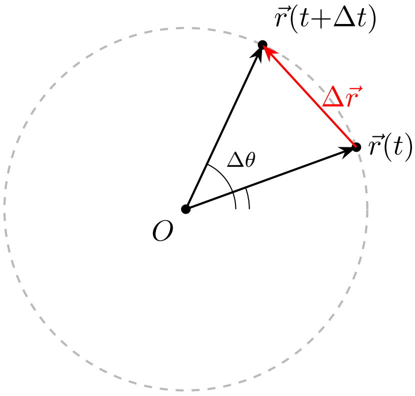


Figure 5.6: Displacement vector $\Delta\vec{r}$ for circular motion over an interval Δt.

Tangential and Radial Acceleration

In polar coordinates, the acceleration has two components: a tangential component a_θ and a radial component a_r,

$$\vec{a}(t) = a_r\,\hat{r}(t) + a_\theta\,\hat{\theta}(t). \tag{53}$$

The unit vectors $\hat{r}(t)$ and $\hat{\theta}(t)$ change direction with time and are therefore not constant. To compute the acceleration, we differentiate the velocity $\vec{v}(t) = r(d\theta/dt)\,\hat{\theta}(t)$ using the product rule,

$$\vec{a}(t) = r\frac{d^2\theta}{dt^2}\,\hat{\theta}(t) + r\frac{d\theta}{dt}\frac{d\hat{\theta}}{dt}. \tag{54}$$

Substituting $d\hat{\theta}/dt = -(d\theta/dt)\hat{r}(t)$,

$$\vec{a}(t) = r\frac{d^2\theta}{dt^2}\,\hat{\theta}(t) - r\left(\frac{d\theta}{dt}\right)^2\hat{r}(t). \tag{55}$$

The two components are therefore

$$a_\theta = r\frac{d^2\theta}{dt^2}, \tag{56}$$

$$a_r = -r\left(\frac{d\theta}{dt}\right)^2 = -r\omega^2 < 0. \tag{57}$$

Because $a_r < 0$, the radial component $a_r(t) = -r\omega^2\,\hat{r}(t)$ is always directed towards the centre of the circular orbit. This is the centripetal acceleration.

Circular Motion Kinematics A particle moves in a circle of radius R. At $t = 0$ it is on the x-axis. The angle with the positive x-axis is $\theta(t) = At^3 - Bt$, where A and B are positive constants. Find (a) the velocity vector and (b) the acceleration vector in polar coordinates. At what time is the centripetal acceleration zero?

Solution. The derivatives are $d\theta/dt = 3At^2 - B$ and $d^2\theta/dt^2 = 6At$. The velocity vector is

$$\vec{v}(t) = R\frac{d\theta}{dt}\,\hat{\theta}(t) = R(3At^2 - B)\,\hat{\theta}(t).$$

The acceleration vector is

$$\vec{a}(t) = R\frac{d^2\theta}{dt^2}\,\hat{\theta}(t) - R\left(\frac{d\theta}{dt}\right)^2\hat{r}(t) = 6RAt\,\hat{\theta}(t) - R(3At^2 - B)^2\hat{r}(t).$$

The centripetal acceleration is zero when $d\theta/dt = 0$, that is, when

$$3At_1^2 - B = 0 \implies t_1 = \sqrt{\frac{B}{3A}}.$$

Period and Frequency for Uniform Circular Motion

If the total tangential force on the object is zero, the tangential acceleration vanishes,

$$a_\theta = 0, \tag{58}$$

and the speed remains constant. This is *uniform circular motion.* The acceleration reduces to

$$\vec{a}_r(t) = -r\omega^2\,\hat{r}(t) \qquad \text{(uniform circular motion).} \tag{59}$$

Because the speed $v = r\omega$ is constant, the time to complete one full revolution, the *period* T, is constant. In one period the object travels the circumference $2\pi r$, so

$$2\pi r = vT \implies T = \frac{2\pi r}{v} = \frac{2\pi r}{r\omega} = \frac{2\pi}{\omega}. \tag{60}$$

The *frequency* f is the reciprocal of the period,

$$f = \frac{1}{T} = \frac{\omega}{2\pi}. \tag{61}$$

The SI unit of frequency is the hertz: $[\mathrm{Hz}] = [\mathrm{s}^{-1}]$. The magnitude of the centripetal acceleration can be written in several equivalent forms. Using $v = r\omega$,

$$|a_r| = r\omega^2, \tag{62}$$

$$|a_r| = \frac{v^2}{r}, \tag{63}$$

$$|a_r| = 4\pi^2 r f^2, \tag{64}$$

$$|a_r| = \frac{4\pi^2 r}{T^2}. \tag{65}$$

Which form to use depends on what information is given about the orbit.

Geometric Interpretation of the Radial Acceleration

An object in circular orbit is always accelerating towards the centre. The velocity is always tangent to the circle, so its direction is continuously changing as the object moves. Because the velocity changes direction, the acceleration is non-zero even when the speed is constant.

The magnitude of the change in velocity is $|\Delta\vec{v}| = 2v\sin(\Delta\theta/2) \approx v\,\Delta\theta$ for small $\Delta\theta$ (Figure 5.7). The magnitude of the radial acceleration is therefore

$$|a_r| = \lim_{\Delta t\to 0}\frac{|\Delta\vec{v}|}{\Delta t} = v\lim_{\Delta t\to 0}\frac{\Delta\theta}{\Delta t} = v\frac{d\theta}{dt} = v\omega = \frac{v^2}{r}. \tag{66}$$

As $\Delta\theta \to 0$, the direction of $\Delta\vec{v}$ is perpendicular to $\vec{v}$ and directed inward, in the $-\hat{r}$-direction, confirming the centripetal character of the acceleration.

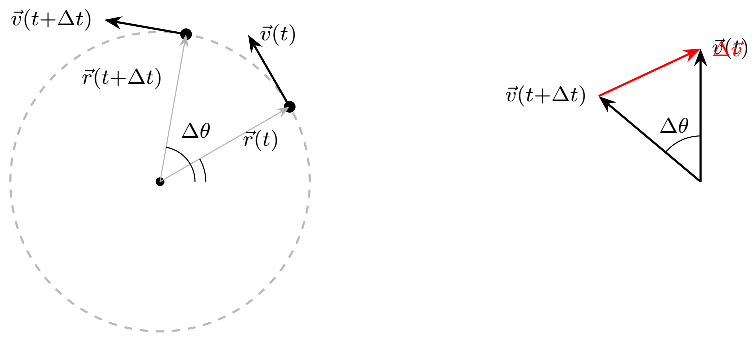


Figure 5.7: Left: velocity vectors at two nearby points on a circular orbit. Right: the change in velocity $\Delta\vec{v}$, which points inward as $\Delta\theta \to 0$.

Angular Velocity and Angular Acceleration

Angular Velocity

We always choose a right-handed cylindrical coordinate system. With the positive z-axis pointing up, the angle θ increases counterclockwise as viewed from above. The *angular velocity vector* for a point object undergoing circular motion about the z-axis is directed along the z-axis,

$$\vec{\omega} = \frac{d\theta}{dt}\,\hat{k} = \omega_z\,\hat{k}. \tag{67}$$

The angular speed is the magnitude of the z-component, $\omega \equiv |\omega_z| = |d\theta/dt|$. If the object rotates counterclockwise, $\omega_z = d\theta/dt > 0$ and $\vec{\omega}$ points in the $+\hat{k}$-direction. If the object rotates clockwise, $\omega_z < 0$ and $\vec{\omega}$ points in the $-\hat{k}$-direction.

The velocity and angular velocity are related by the cross product,

$$\vec{v} = \vec{\omega} \times \vec{r} = \frac{d\theta}{dt}\,\hat{k} \times r\hat{r} = r\frac{d\theta}{dt}\,\hat{\theta}. \tag{68}$$

Angular Velocity A particle moves in a circle of radius R. At $t = 0$ it is on the x-axis. The angle with the positive x-axis is $\theta(t) = At - Bt^3$, where A and B are positive constants. Find (a) the angular velocity vector, (b) the velocity vector, (c) the time t_1 when the angular velocity is zero, and (d) the direction of $\vec{\omega}$ for $t < t_1$ and $t > t_1$.

Solution. The derivative of $\theta(t)$ is

$$\frac{d\theta}{dt} = A - 3Bt^2.$$

(a) The angular velocity vector is

$$\vec{\omega}(t) = (A - 3Bt^2)\,\hat{k}.$$

(b) The velocity is

$$\vec{v}(t) = R(A - 3Bt^2)\,\hat{\theta}(t).$$

(c) The angular velocity is zero when

$$A - 3Bt_1^2 = 0 \implies t_1 = \sqrt{\frac{A}{3B}}.$$

(d) For $t < t_1$, $d\theta/dt > 0$ so $\vec{\omega}$ points in the $+\hat{k}$-direction (counterclockwise). For $t > t_1$, $d\theta/dt < 0$ so $\vec{\omega}$ points in the $-\hat{k}$-direction (clockwise).

Angular Acceleration

The *angular acceleration vector* for circular motion about the z-axis is

$$\vec{\alpha} = \frac{d^2\theta}{dt^2}\,\hat{k} = \alpha_z\,\hat{k}. \tag{69}$$

The SI units of angular acceleration are $[\mathrm{rad}\cdot\mathrm{s}^{-2}]$. There are four combinations of the signs of ω_z and α_z to consider.

When $\vec{\alpha}$ points in the $+\hat{k}$-direction ($\alpha_z > 0$): (i) if $\omega_z > 0$ (counterclockwise), the object is speeding up counterclockwise; (ii) if $\omega_z < 0$ (clockwise), the object is slowing down.

When $\vec{\alpha}$ points in the $-\hat{k}$-direction ($\alpha_z < 0$): (iii) if $\omega_z > 0$ (counterclockwise), the object is slowing down; (iv) if $\omega_z < 0$ (clockwise), the object is speeding up clockwise.

The tangential and radial components of the acceleration (Eqs. (56) and (57)) can be written compactly as $a_\theta = r\alpha_z$ and $a_r = -r\omega_z^2$.

Integration and Circular Motion Kinematics A point object is constrained to travel in a circle. The z-component of the angular acceleration is

$$\alpha_z(t) = \begin{cases} b\left(1 - \dfrac{t}{t_1}\right), & 0 \le t \le t_1, \\ 0, & t > t_1, \end{cases}$$

where $b > 0$ has units $\mathrm{rad}\cdot\mathrm{s}^{-2}$. (a) Find the angular velocity at $t = t_1$. (b) Through what angle has the object rotated at $t = t_1$?

Solution. (a) Starting from rest ($\omega_z(0) = 0$), we integrate:

$$\omega_z(t_1) = \int_0^{t_1} b\left(1 - \frac{t'}{t_1}\right) dt' = b\left[t_1 - \frac{t_1}{2}\right] = \frac{bt_1}{2}.$$

(b) First find $\omega_z(t)$ for $0 \le t \le t_1$:

$$\omega_z(t) = \int_0^{t} b\left(1 - \frac{t'}{t_1}\right) dt' = b\left(t - \frac{t^2}{2t_1}\right).$$

Then integrate again to find the angle:

$$\theta(t_1) - \theta(0) = \int_0^{t_1} \omega_z(t')\,dt' = \int_0^{t_1} b\left(t' - \frac{t'^2}{2t_1}\right) dt' = b\left[\frac{t_1^2}{2} - \frac{t_1^2}{6}\right] = \frac{bt_1^2}{3}.$$

CHAPTER 6